GOURAB SINGHA

CS UNDERGRADUATE STUDENT

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Github • Linkedin

CARRIER OBJECTIV

Passionate and detail-oriented Computer Science graduate with strong proficiency in Python, SQL, and machine learning frameworks. Experienced in building AI-based models for classification, recommendation, and data analysis, emphasizing optimization and scalability. Possesses a solid foundation in cybersecurity practices, including web penetration testing and vulnerability assessment, ensuring robust and secure solutions. Committed to contributing technical innovation, analytical insight, and problem-solving excellence to organizational success.

PROFESSIONAL EXPERIENCE

Intern | AI/ML

Zensoft Lab, Dhaka | October 2025 – Ongoing **website**

- Analyzed and evaluated open-source large language models (LLMs) to identify suitable architectures for specific applications.
- Fine-tuned LLMs for domain-specific tasks using Python, Hugging Face Transformers, and PyTorch frameworks.
- Collaborated with team members to improve workflow efficiency and optimize training pipelines.
- Documented experiment results and contributed insights for model performance improvement and scalability.

SKILLS

- **Language:** Python, SQL(MySQL, PostgreSQL), C , HTML/CSS , Java, JavaScript, PHP
- **Competitive Programming:** Bee-Crowd, Leet code, Code forces
- **Tools:** Kaggle, Google Colab, Git, GitHub, Vs code, Pycharm, AutoCAD
- **Penetration Testing:** Web pentesting (OWASP Top10), Vulnerability assessment, Exploitation, tools: Burp, Nmap, Metasploit, SQLmap, WP Scan
- **Typing Speed:** 100wpm+ (Monkeytype)
- **Operating System:** Linux, Windows 7/10/11
- **Soft Skills:** Analytical thinking , Independent research ability, Project Leadership, Technical Communication, Team Collaboration, Fluent in English

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EDUCATION

Green University Of Bangladesh

- BSC in CSE | 3.3 out 4 | 2022 - Ongoing

Narail Govt. Victoria College

- HSC | Science | 4.17 out 5 | 2017 - 2019

PROJECT

Deep Q-Learning for Employee Performance Classification with Human-in-the-Loop Feedback | 4th Year Q3 **github**

- Developed a Deep Q-Learning model for automated employee performance evaluation using behavioral metrics.
- Implemented a human-in-the-loop feedback system for continuous model improvement through incremental training.

- Engineered a data pipeline to extract six key features from attendance records, including work hours, punctuality, and consistency.
- Achieved adaptive learning with experience replay and epsilon-greedy exploration.
- **Tools & Technologies:** Python, PyTorch, Pandas, NumPy, Deep Q-Network (DQN), Reinforcement Learning.

Brain Tumor Detection from MRI image using Machine Learning | 4th Year Q1 **github**

- Developed a ML model for automated tumor detection using feature extraction and classification
- Enhanced diagnostic accuracy through data preprocessing and performance optimization
- **Tools & Technologies:** Python, Pands, NumPy, Scikit-learn, Matplotlib, TensorFlow

Hybrid Movie Recommendation System using Machine Learning | 3rd Year Q2 **github**

- Developed a hybrid recommendation system using content-based and KNN-based collaborative filtering
- Integrated MySQL-based user authentication and personalized movie suggestions via Streamlit frontend
- Improved recommendation accuracy through data preprocessing and ML integration
- **Tools & Technologies:** Python, Pandas, NumPy, Scikit-learn, Streamlit, MySQL, NearestNeighbors

Custom Encryption Tool (AES + Rail Fence Cipher) | 3rd Year Q1 **github**

- Developed a hybrid encryption tool combining AES and Rail Fence Cipher for secure text and PDF encryption/decryption
- Designed a user-friendly GUI with options for manual input, file upload and encrypted PDF generation
- **Tools & Technologies:** Python, Tkinter, PyPDF2, ReportLab, PyCryptodome(AES)

E-Commerce Platform | 2rd Year Q3 **github**

- Developed a full-stack e-commerce website with product listing, user authentication, shopping cart and order management
- Focused on seamless user experience and secure transaction flow
- **Tools & Technologies:** PHP, MySQL, HTML, CSS, JavaScript

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RESEARCH

3D Deep Learning Model for Enhanced Detection of Brain Tumor from MRI and PET Images - Ongoing

- Designing a 3D deep learning model for multi-modal MRI and PET image analysis to improve tumor segmentation accuracy
- Performing data preprocessing, normalization and 3D volume reconstruction for effective model input
- Investigating hybrid loss functions and performance metrics (Dice, IoU) for robust evaluation
- Aiming to enhance diagnostic reliability and reduce manual annotation effort for radiologists

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CERTIFICATES

Python Training | 2024-2025

Institute of Information Technology (IIT), Jahangirnagar University | EDGE Project, Bangladesh Computer Council, ICT Division

CyberSecurity and Ethical Hacking | 2024

Codemanbd

INTERESTS

- Machine Learning & AI projects, building and fine -tuning ML models
- Exploring new technologies and solving challenging problems